

An introduction to programming concepts with Scratch



What can we do with Scratch?

- Learn some computing concepts.
- Learn some practical algorithms.
- Use Scratch as computing tool.
- Have fun with Scratch creating stories, games, art.

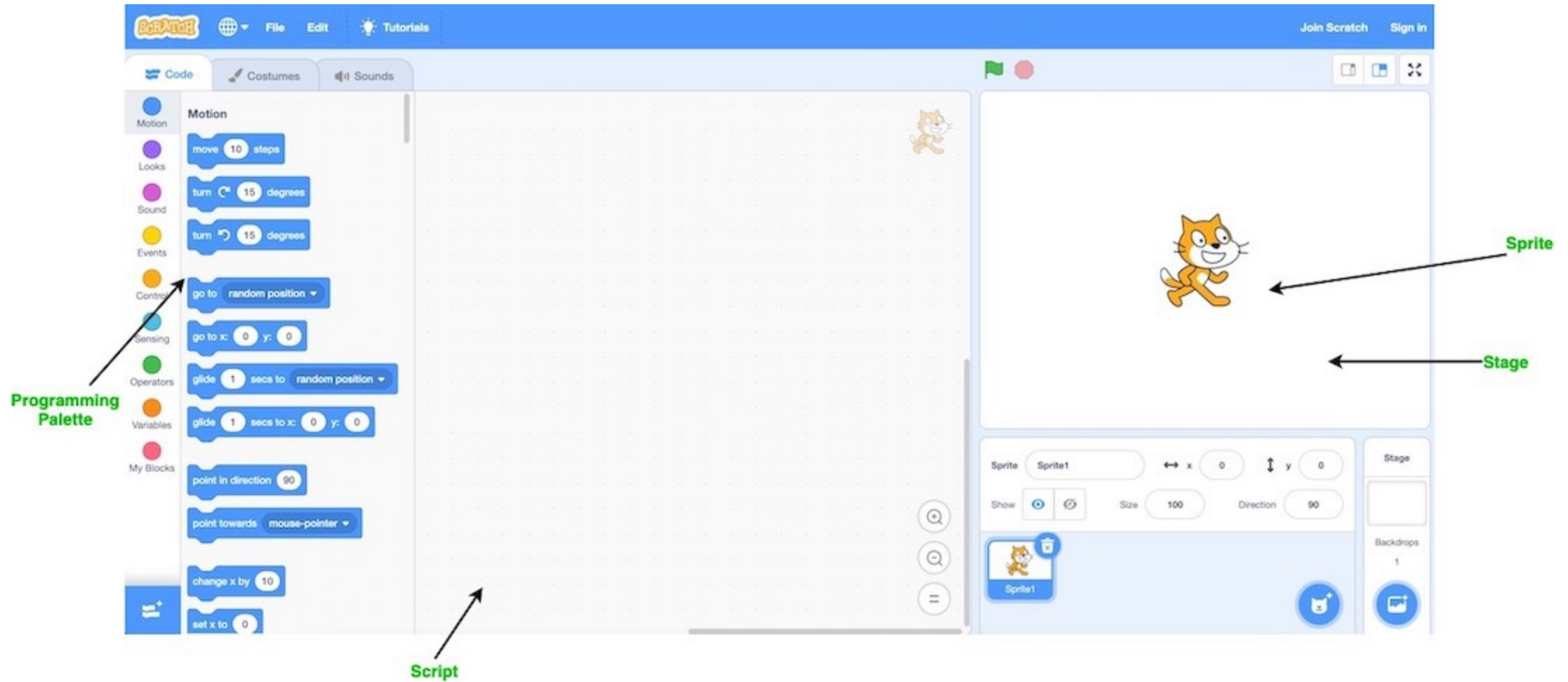
Part I: Looks and Motion

- Goals:
 - Learn Scratch programming environment
 - Learn the looks and motion menus
 - Write a sequence of instructions

Exercise (10 min)

- 1 Create your own scratch account
- 2 Find out -What is a Remix in scratch

Explore Scratch





Find på

Udforsk

Idéer

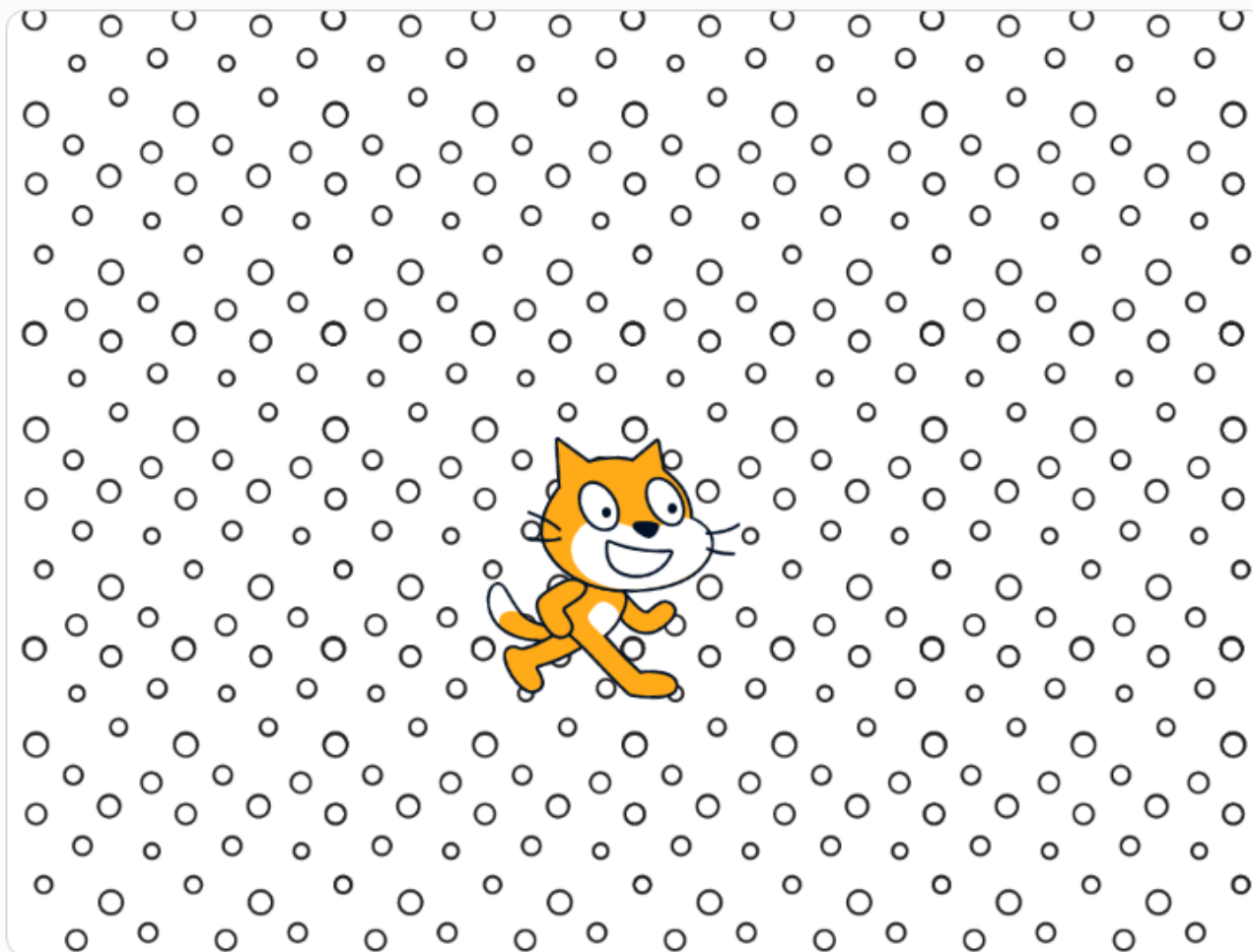
Om

Søg



shirleybee

Untitled



Instruktioner

Tell people how to use your project (such as which keys to press).

Noter og anerkendelser

How did you make this project? Did you use ideas, scripts or artwork from other people? Thank them here.

Start scratch

- Click on the cat icon
- Scratch programming environment comes up
- We will first do simple things



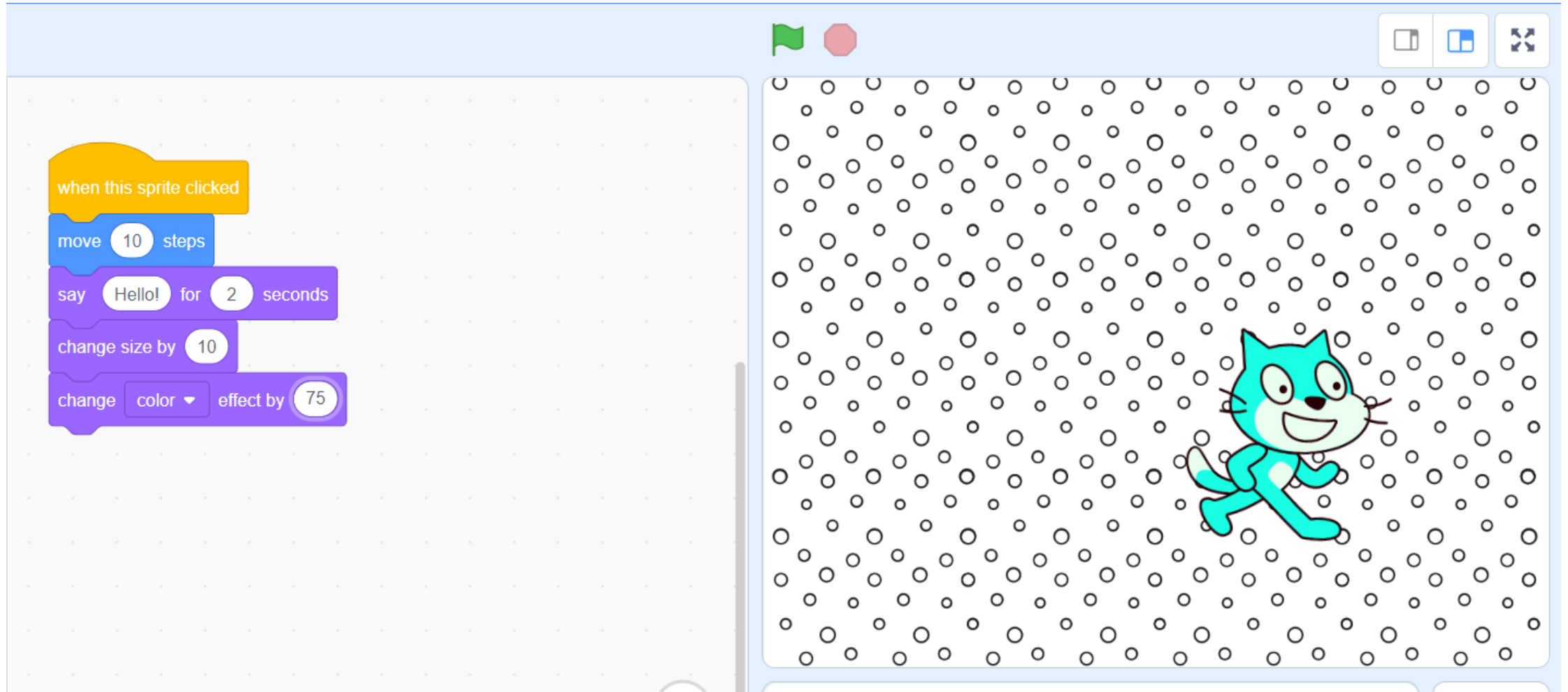
Our Scratch environment

- ***Stage*** is at upper right (where actors act and information is displayed)
- ***Sprite*** is another name for actor.
- Instruction menus/panel at left (instructions for the actors)
- A ***script*** is another name for program or method; a script tells the actor what to do.
- Programming area in center; here is where we construct scripts for the sprites

We start with the cat sprite

We will learn to use many other sprites later. For now we use the cat as our only actor. We script the BEHAVIOR of our cat.

Our script (program)



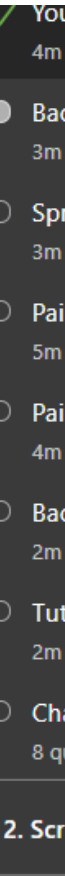
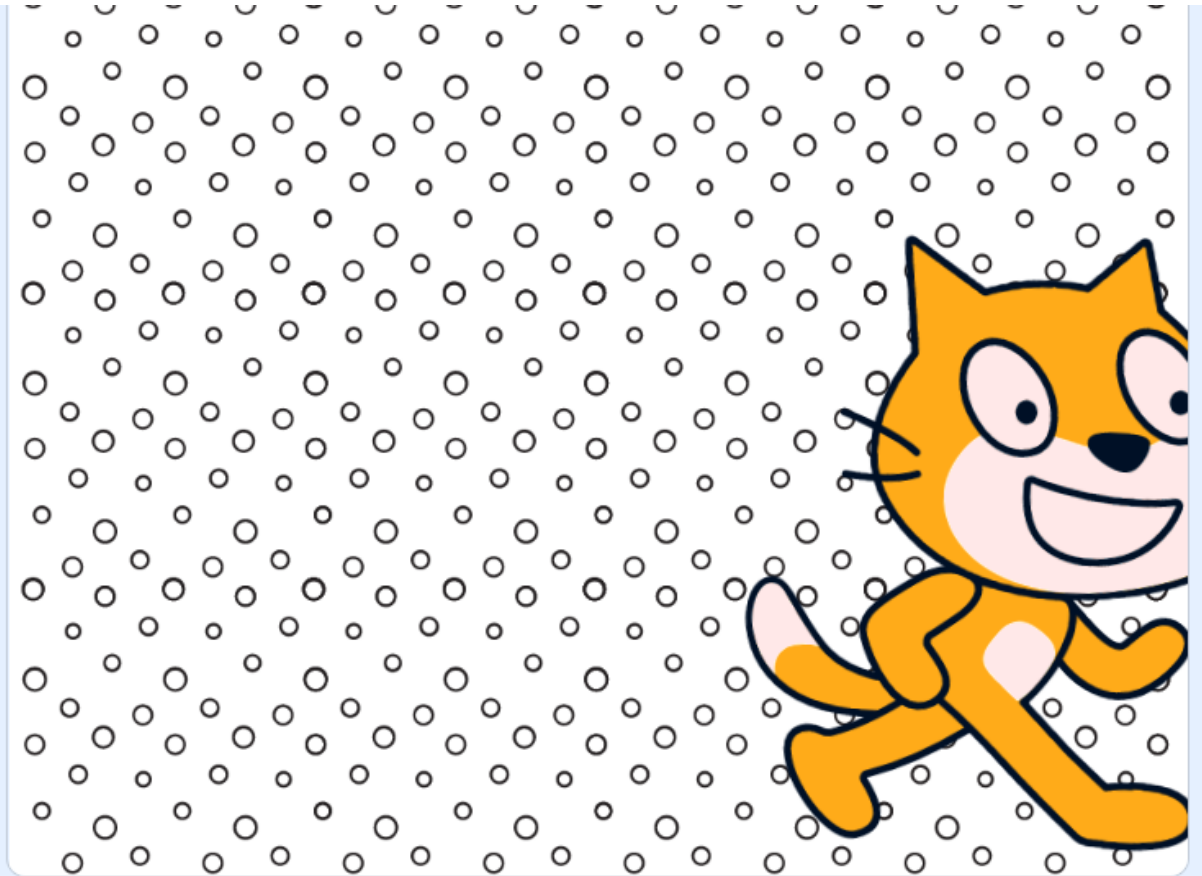
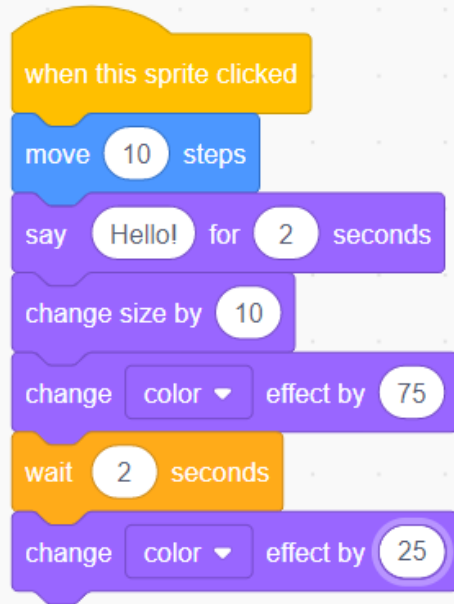
A sequence or block is a simple program

- The first instruction is done first
- The second instruction is done second
- The last instruction is done last.
- (if any one instruction is done, then every one of them is done)

The WAIT instruction

- WAIT is needed to slow down the acting so we can see or hear it properly (computers are too fast sometimes)
- Get the wait instruction from the CONTROL menu. Ignore the other menu options for now.
- Insert a wait in our looks script

2 second pause between changing color



Exercise: write a script to do the following

- When click start
- Double the size of the sprite
- Wait 2 seconds
- Change the color of the sprite to green
- Wait 4 seconds
- Change the whirl effect to 100
- Say “That’s all folks!”

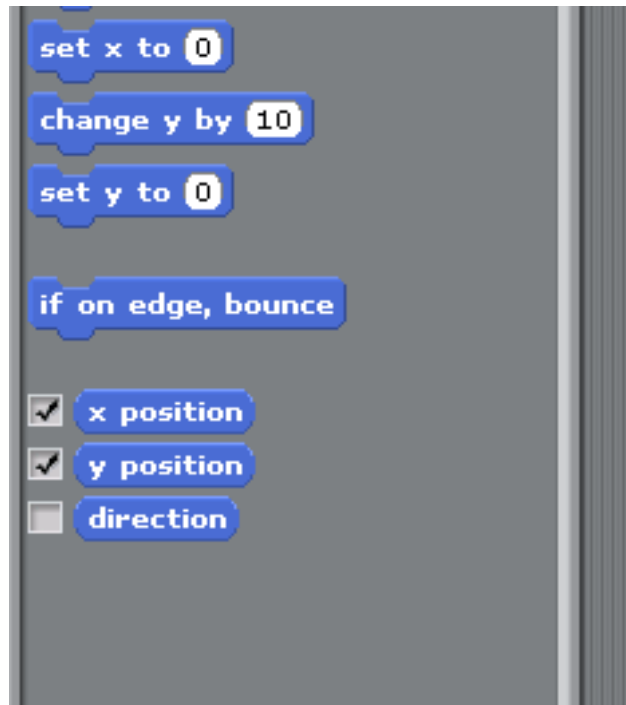
https://en.scratch-wiki.info/wiki/Graphic_Effect

The MOTION menu

How to locate and orient a sprite;
moving a sprite.

Position on the stage

- Click on the **Motion** menu.
- Click to check the box for x-position and y-position
- Drag your cat around and note its x-y position.



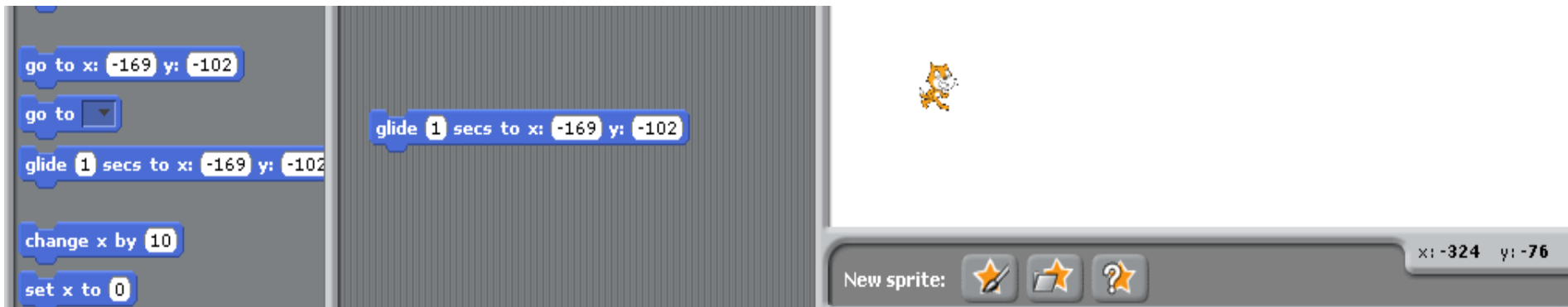
Exercises: goto instruction

- In the Motion menu, drag the “goto xy” instruction to the script panel.
- Edit the *coordinates* and click to see the sprite's position
- A) goto x=200, y=0
- B) goto x=-200, y=0
- C) goto x=200, y=-100

Where
does the
cat go?

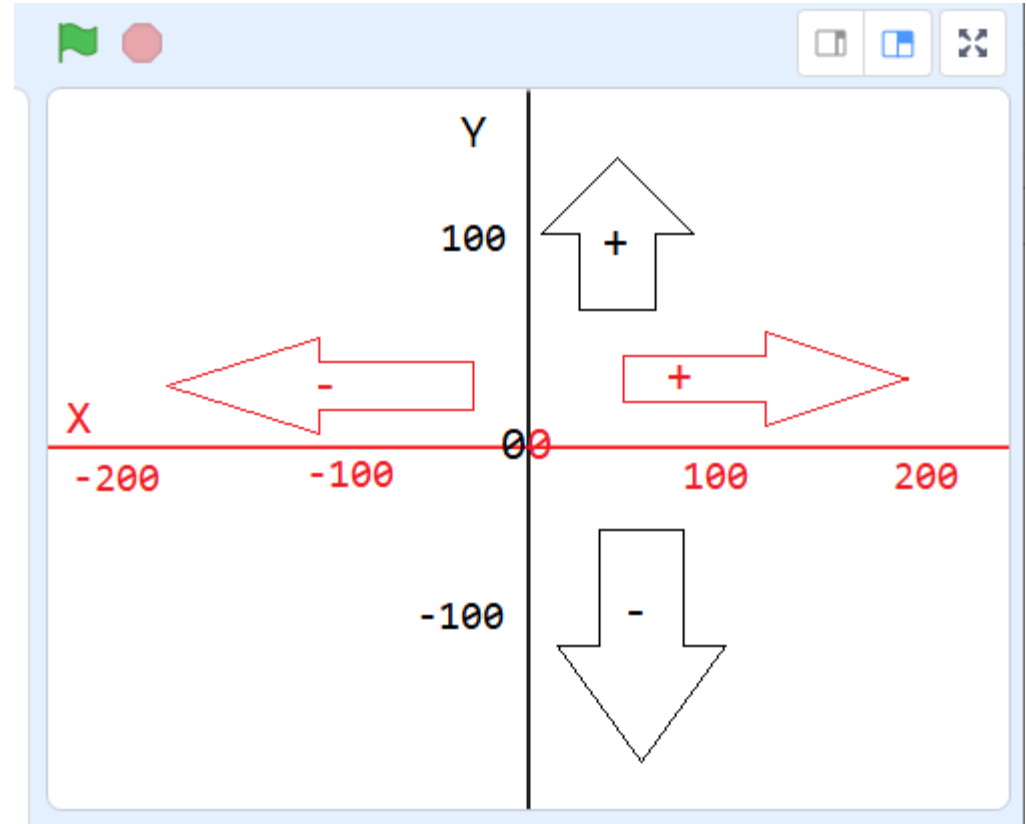
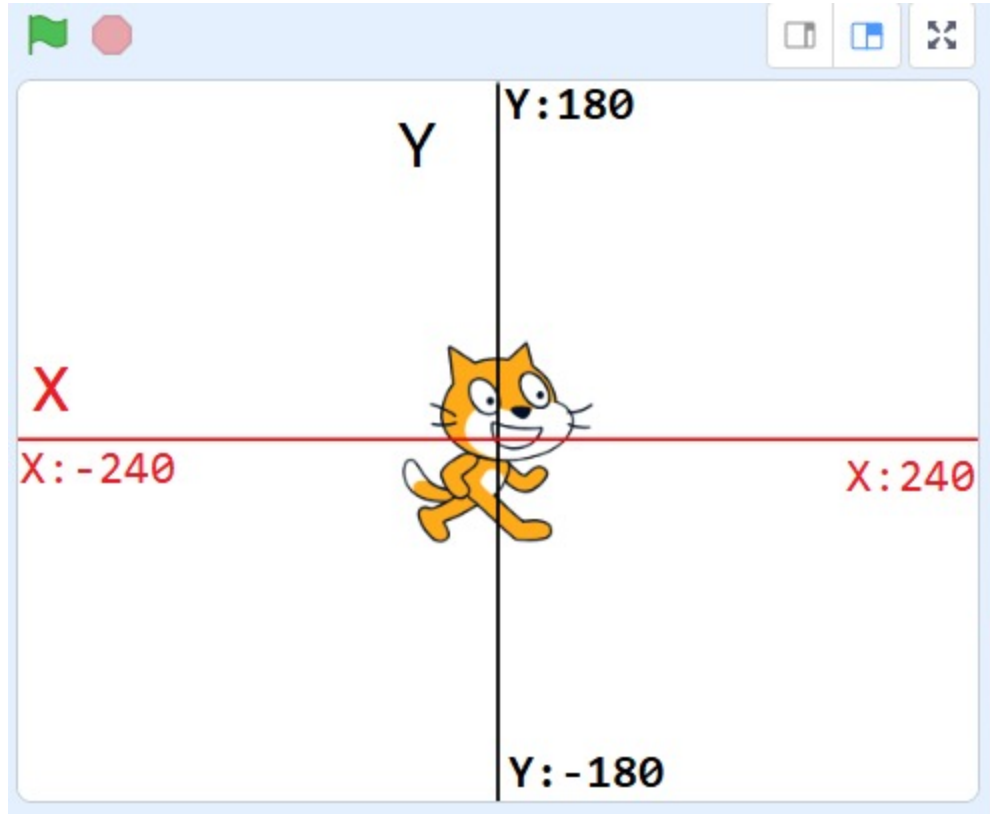
The “glide-to” instruction

- Drag the “glide-to” instruction into your script panel.
- Edit the coordinate values and click to see where your sprites goes.



Angles and directions

Making your sprite go
this way and that way



End of Part I: Outcomes

Student should be able to control the looks of a sprite and its location on the stage. Exercise 1

Part 2: Repetition and Variation

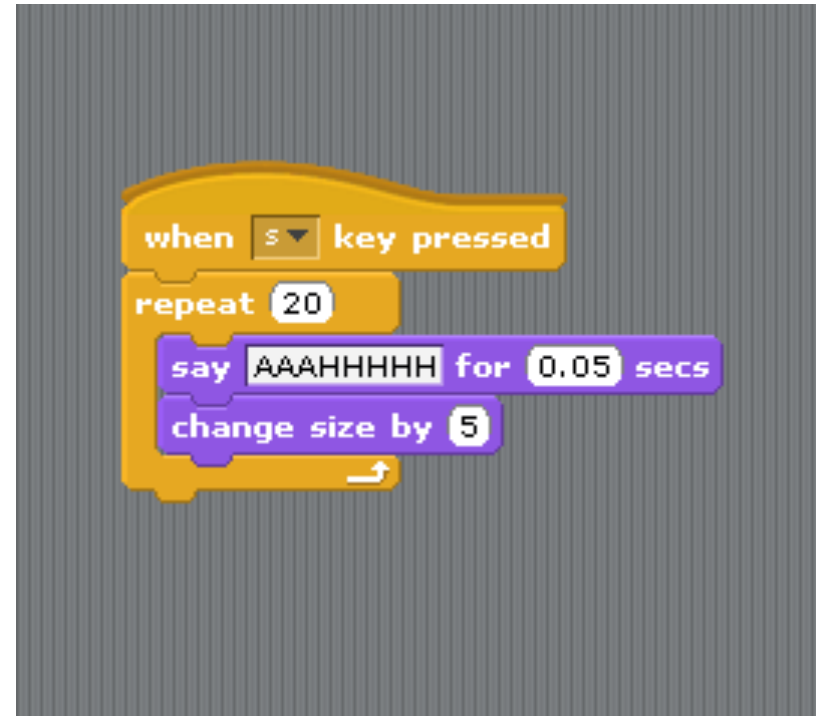
- Goals:
 - Learn how to program with ***repetition***
 - Learn how to change ***costumes***
 - Learn to control execution (behavior) using special keys

The sneezing cat script

- We repeat 20 times

Say “AAAHHHH”
for a short time

Then grow the
cat 5 bigger



Exercise: Bring the cat size down to normal

Repeat 20 times

Briefly say

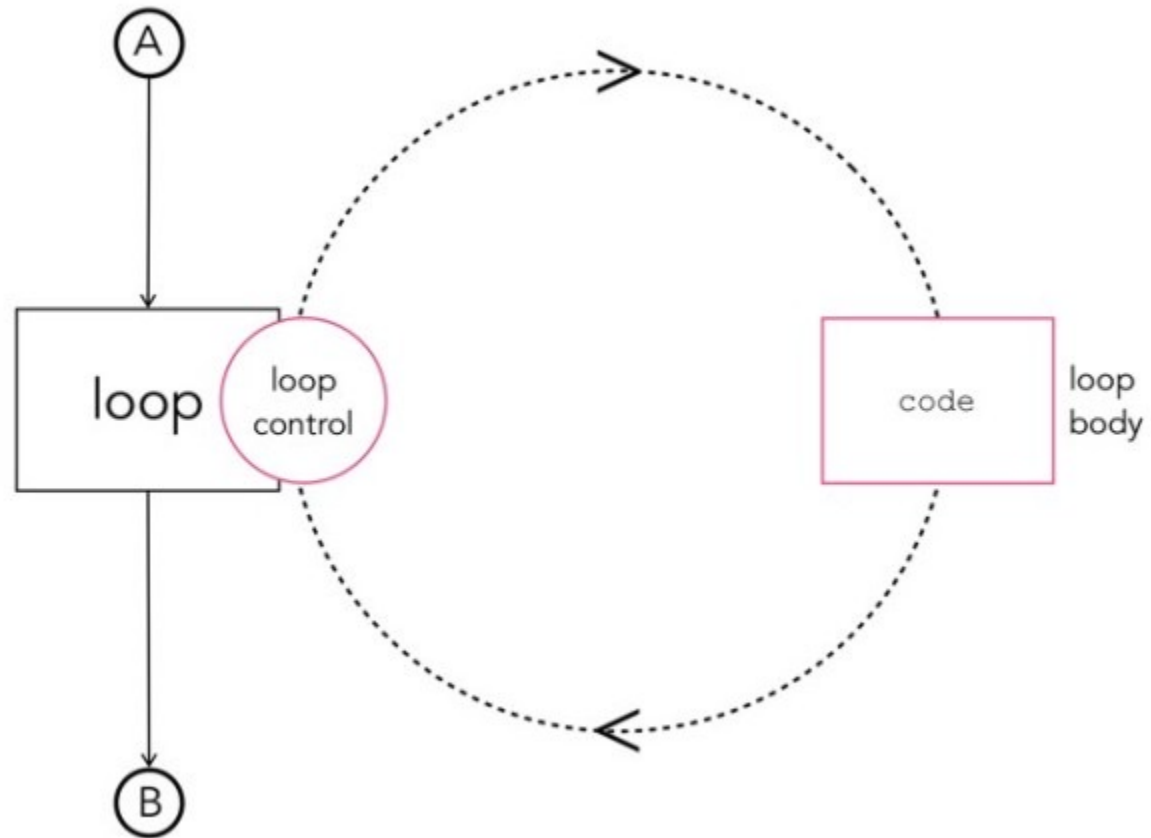
“CCHHEEEW”

Then shrink the

cat’s size by 5

Repeat N times
and repeat
forever are in the
Control menu

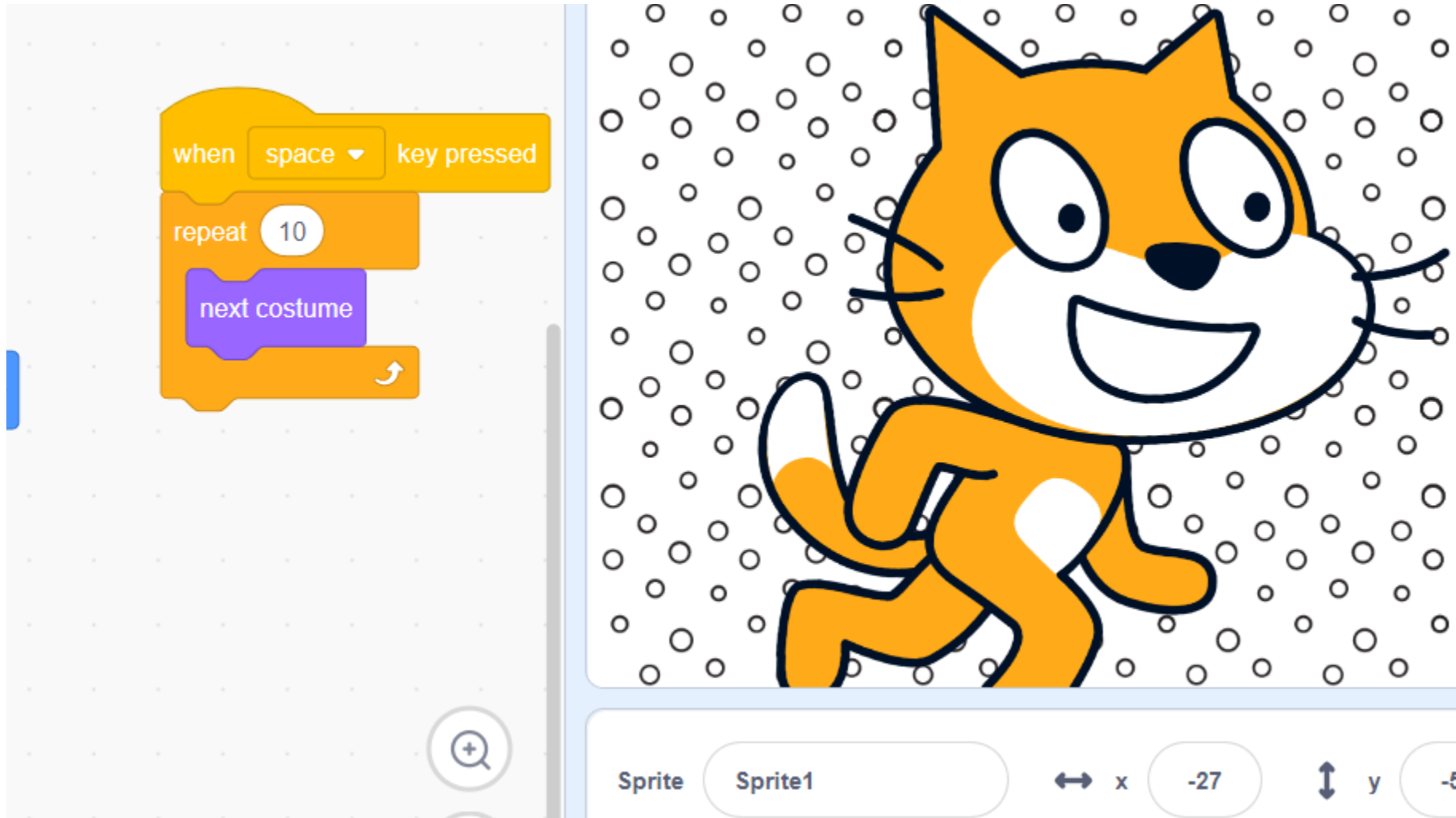
Repeat in programming



Changing costumes

Another way to change the
Appearance of a sprite.

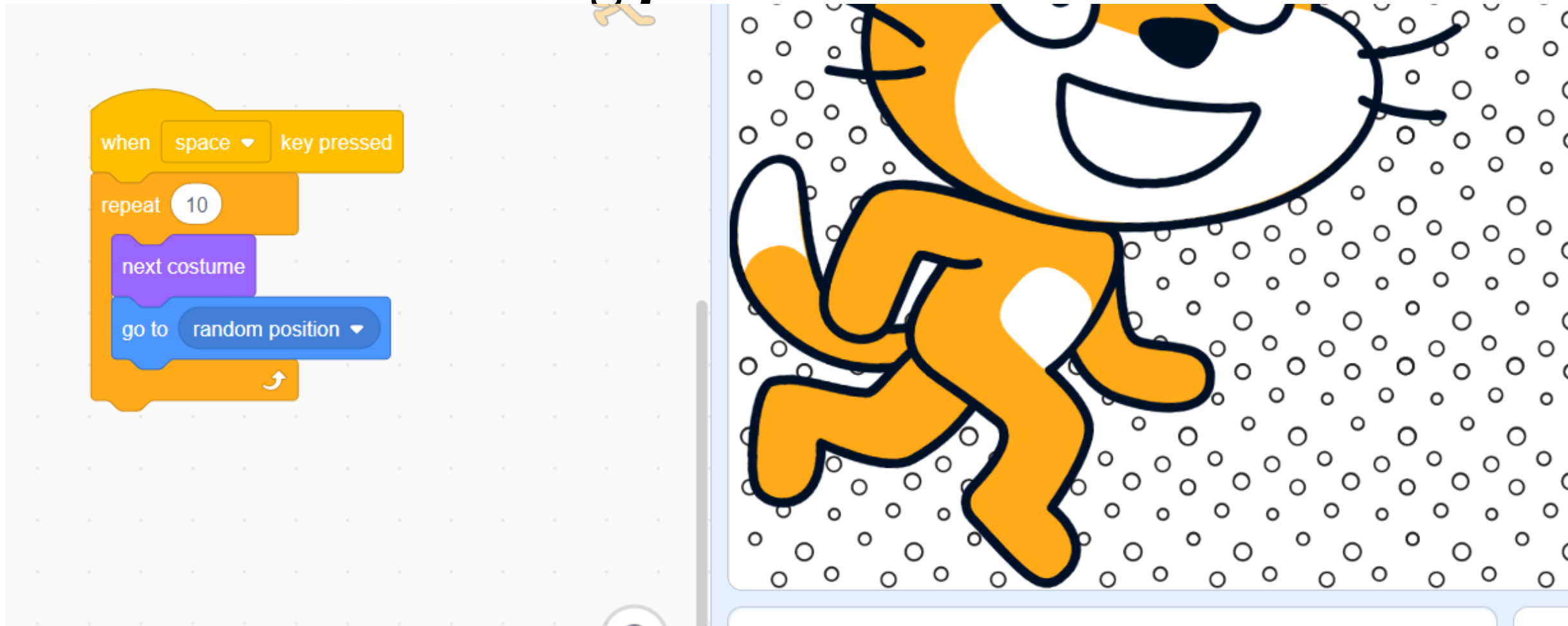
New costume



Changing coordinates

- We can *randomly* set the location of the bat so it will “flutter”.
- Set X to a random number
- Set Y to a random number
- Move the bat to location (X,Y)
- Of course, the bat should remain on stage!

Using ***pick random***



Click the stop sign at the stage upper right to stop the forever loop.

Do we need a wait in the loop? Try it to see the changes

Controlling the direction of a new Sprite

- Make a new Sprite
- Let's create **scripts**.
- Click space bar for random moves
- Click right arrow to move right
- Click left arrow to move left

Exercise Change Sprite

◀ Back

Choose a Sprite

🔍 Search

All

Animals

People

Fantasy

Dance

Music

Sports

Food

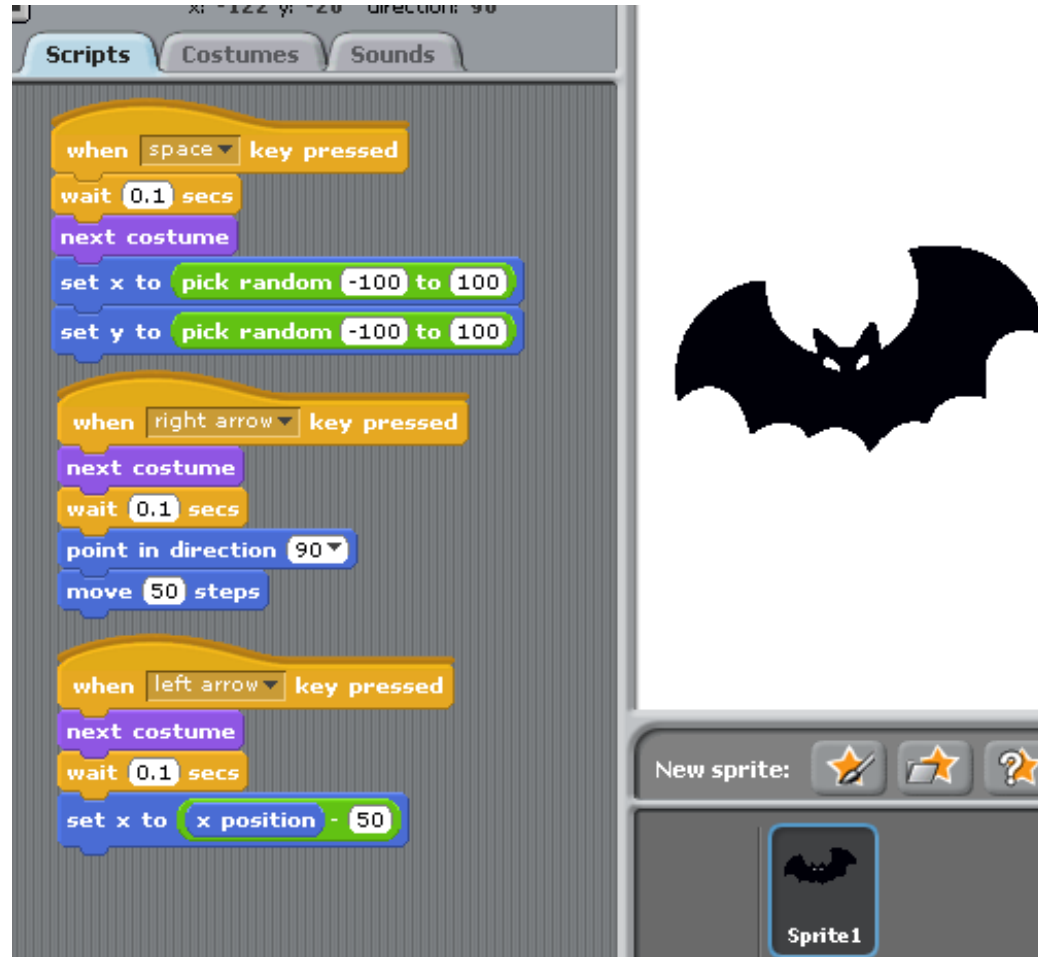
Fashion

Letters

 Abby	 Amon	 Andie	 Anina Dance	 Apple	 Arrow1	 Avery
 Avery Walk...	 Ball	 Ballerina	 Balloon1	 Bananas	 Baseball	 Basketball
						

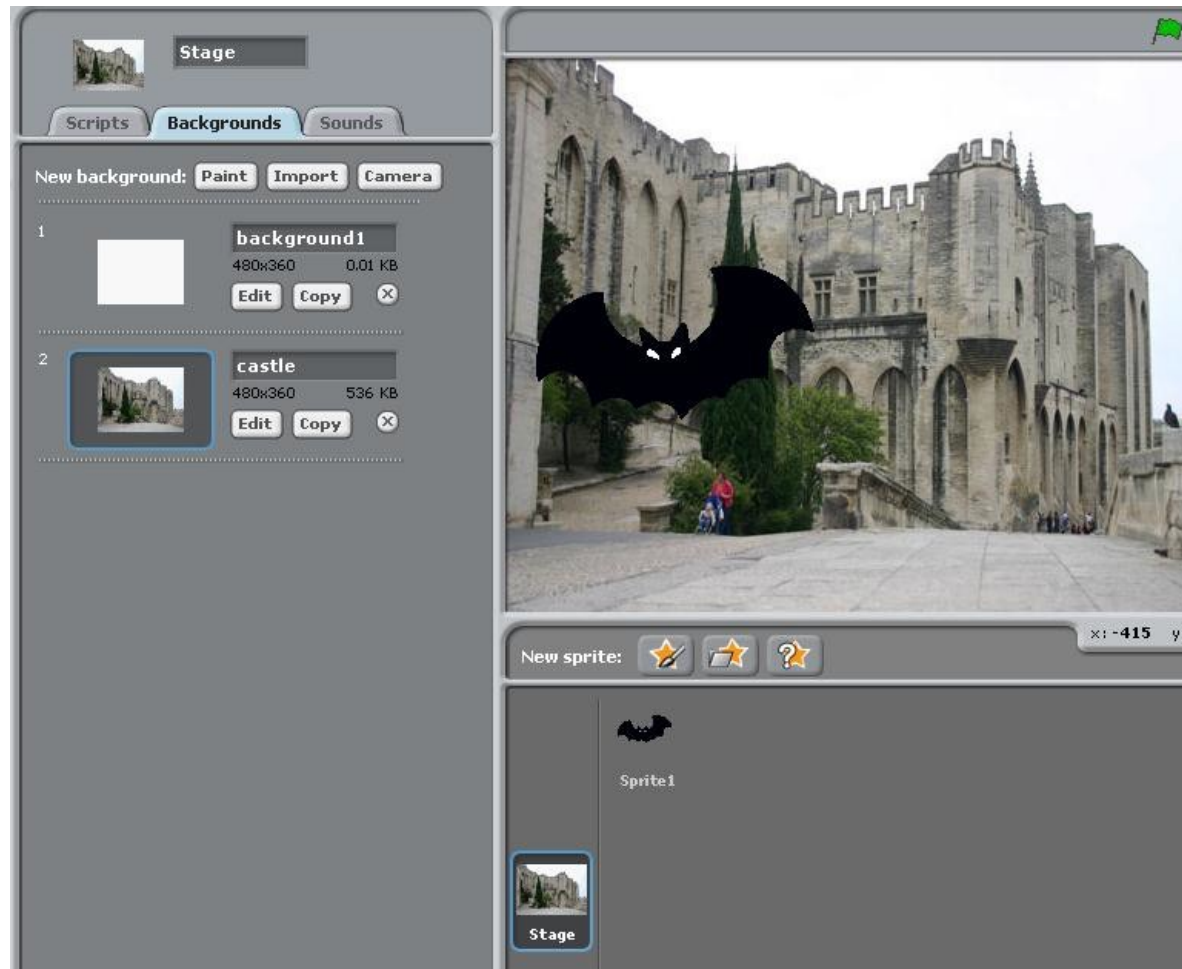
Multiple interactive bat flight

3 different behaviors for the bat



Adding a *background*

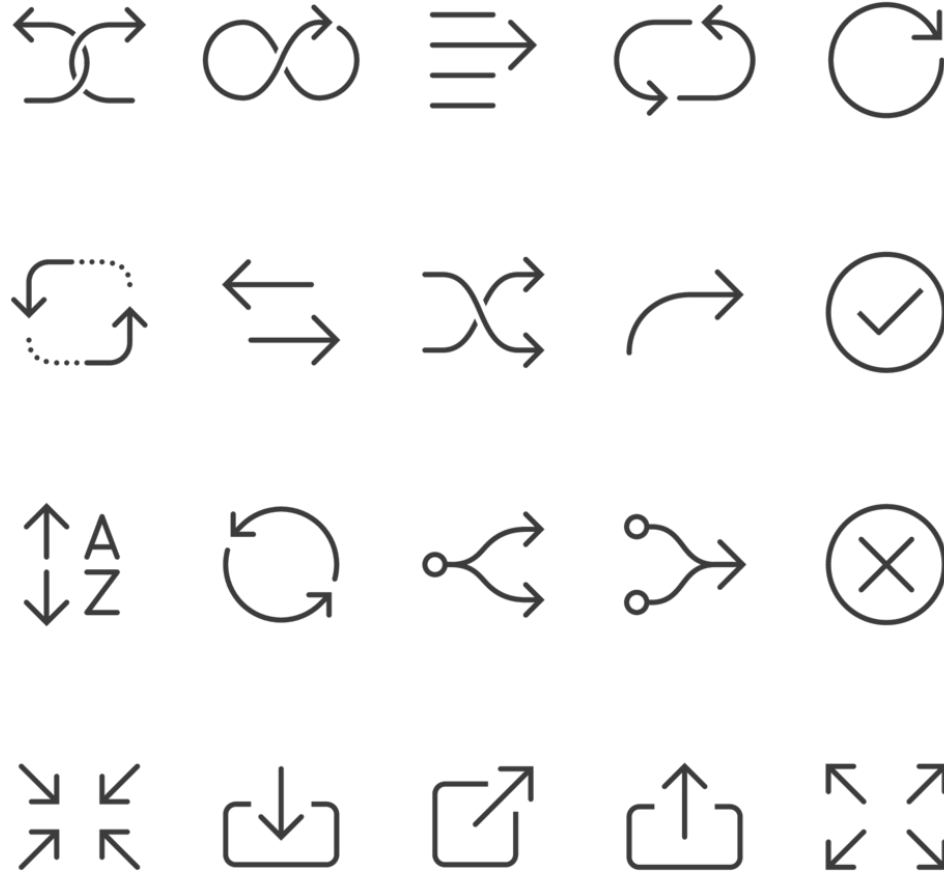
- Click on the ***Stage*** icon at lower right
- Click on ***Backgrounds***
- Click on bat icon (Sprite1)
- Click on ***Scripts***
- FLY AGAIN!



Exercise:

- Add a behavior (script) to move a sprite up with the up arrow.
- Add a behavior to move the sprite down with the down arrow.
- Add more behavior ...

Part : control statement-if





```
when green flag clicked
ask "What's your age?" and wait
say answer
set age to answer
if age < 20 then
say "A boy's will is the wind's will, And the thoughts of youth are long"
```

age 12

answer 12

